Carleton

STAT230: APPLIED REGRESSION ANALYSIS

Spring 2026

MEETINGS:	MWF 5a	CMC 306
	1:50-3 MW; 2:20-3:20 F	
PROFESSOR:	Amanda Luby aluby@carleton	CMC 223
DROP-IN:	Mon 3-4pm	CMC 307
	Tues 2-3	
	Wed 9:30-10:30 (120 priority)	
	Fri 10-11 (230 priority)	
	Or by appointment!	
WEBSITE:	moodle.carleton.edu	
TEXT:	Regression Analysis by Example by A. Hadi | | Freely available through the Library |	
SOFTWARE:	maize.mathcs.carleton.edu (R code)	
	gradescope (homework)	

COURSE DESCRIPTION

Our world is awash in data. To make decisions based on evidence, we must be able to make sense of the data around us and communicate our findings. In this course, we will explore how to incorporate multiple predictors into regression models to answer complex questions. In addition to modeling continuous response variables, we will also explore models for binary data, and binomial and Poisson counts. This course will emphasize model building, model validation, and how to communicate results. We will make frequent use of R.

COURSE OBJECTIVES

After completing this course, you should be able to demonstrate your competency in each of the following areas:

- Develop research questions that can be answered using a regression model
- Fit linear, logistic, and Poisson regression models in R and interpret the output.
- Conduct inference for the model parameters and/or predictions and interpret the results in context.
- Comment on the adequacy of a model for addressing a given question of interest by assessing the assumptions underlying the model.
- Apply appropriate transformations to improve agreement with the model assumptions.
- Use statistical reasoning to select the most adequate for the question at hand.
- Communicate the results of a regression analysis effectively in written format.

WHAT I EXPECT FROM YOU

- **Come prepared to learn**. Complete background readings and bring your notes, assignments, and laptop with you to class. Be ready to engage in discussion and activities from the first minute of class.
- **Do your own work**. Always give credit when using others' ideas, code, and writing. Review the collaboration policy for each assignment and follow it.
- **Show up and join in**. This is a hands-on class where we will learn by doing.
- **Be a good teammate**. Pull your weight, respect deadlines, ask questions when you're confused, and contribute your perspective.
- **Use technology wisely**. Laptops and phones are great tools, but easy distractions. Recognize when they are helping and when they are hurting your learning.
- **Manage your time and workload**. You will have lots to do during this course. I offer options for flexibility, but you will need to balance this flexibility throughout the term.

WHAT YOU CAN EXPECT FROM ME

- **Be accessible**. I will hold four drop-in hours per week and am glad to set up individual appointments as needed.

- **Timely responses**. I will respond to Ed messages within 24 hours and emails within 48 hours on weekdays (and as soon as I can on weekdays).
- **Prompt feedback**. I will return graded assignments to you within one week
- **Seek your feedback**. I will ask for your feedback at least twice during the term, and I welcome your thoughts at any time. I value candid and constructive comments to help me improve as a teacher
- **Support equitable access**. I will work with you and the Office of Accessibility Resources (OAR) to make sure accommodations are in place so everyone can fully participate in this course.
- **Care about your well-being**. I recognize that life outside the classroom can be stressful. I will do my best to be flexible and supportive, and to connect you with resources when you need them.

COURSE COMPONENTS

MEETINGS

There will be three course meetings per week (Mondays, Wednesdays, and Fridays). Daily attendance is expected. Course meetings will combine demonstrations using R, theoretical background, and in-class group exercises. Please show up ready to participate and engage from the first minute of class!

HOMEWORK ASSIGNMENTS

Homework will be assigned once per week on moodle (typically due Fridays) and should be submitted on gradescope before class. There is an 8 hour grace period for each assignment. If you are ill or have an emergency, please contact me to arrange an alternative due date. Assignment submissions should be neatly written and scanned or submitted as a PDF file. You must show all work to receive credit.

CASE STUDIES

There will be three case studies spaced throughout the term. These will ask you to analyze a dataset using tools from the course and clearly report your findings in writing. These are larger and more open-ended than a typical homework assignment and are preparation for your final project. You'll work in pairs or small groups (which I will form with your input).

EXAMS

A key part of statistical literacy is being able to recall concepts "on the fly". I use in-class exams to assess your understanding of class material without access to resources. There will be three midterm exams — tentatively scheduled for **Week 3 Monday, 6 Wednesday, and 9 Monday**.

If you miss a class when there is an exam with less than 1 week notice, then there are no revisions or makeups offered. If must miss an exam due to an illness or other last minute emergency, please let me know in advance to arrange an alternative.

FINAL PROJECT

The final project is your capstone experience for the course. It's designed to tie everything together that you've learned: you will find a dataset, develop research questions, construct a model, and report your findings. You'll work in pairs or small groups (which I will form with your input). More details and grading criteria will be provided later in the term.

COMMUNICATION

Assignments, note sets, and grades will be posted on Moodle. We will also use Ed for announcements, discussion, and homework questions. Please use Ed for any homework or course content questions; email me privately with any personal matters (grade discussions, illness, emergency, etc.). Any time-sensitive announcements will be posted on Ed. It is your responsibility to make sure that your notification settings allow time-sensitive announcements to reach you.

GRADING POLICIES

Grades are an imperfect measure of learning, and no grading scheme will perfectly capture your "true" learning in this course. This grading scheme is designed to reward you for consistently participating, staying on top of the course material, and demonstrating proficiency with the content through in-class assessments.

Final grades will be calculated as follows:

- 15% Homework
- 30% Exams (10% each)
- 25% Final Project
- 25% Case Studies (8.33% each)
- 5% Attendance and participation in group work

More than 5 absences from class will result in a 0% attendance and participation grade.

Letter grades will be assigned based on the usual grading scale (A = 93% and above, A- = 90-92.9%, B+ = 88-89.9%, B = 83-87.9%, etc.).

A note on the "genius myth": I've found that many carls buy into the "genius myth" when it comes to math/stat courses: that you need to be a "math person" and have some innate mathematical ability in order to do well. This could not be further from the truth! The best statisticians don't necessarily have the "best" math or programming background, but are people that are able to formulate interesting questions and use math and programming to answer those questions. Many of the best statisticians I know became statisticians because they were initially interested in something else (biology, public health, psychology, neuroscience, physics, etc.) and realized that being able to answer important questions with data was not only valuable but fun and interesting. Being able to perform interesting statistical analyses is a skill that is learned, not an innate ability, and working hard at developing that skill is the point of this course.

REGRADE REQUESTS

Grading is often a tedious task, and the grading team will sometimes make mistakes. I am always happy to fix these mistakes, and gradescope makes it easy to do so. Regrade requests must be submitted on gradescope within two weekdays after an assignment or exam has been returned to you. Regrade requests are for administrative errors or obvious grading mistakes. I will not consider regrade requests for anything that applied to the entire class (e.g. "I think this mistake should only be worth 1 point instead of 2" or "I didn't realize we had to do X"). If you submit two or more inappropriate regrade requests, I will not consider additional regrade requests from you for the remainder of the term. If you're unsure whether you should file a regrade request or not, just ask! You are always welcome to discuss any grading questions with me in office hours.

TOKENS

You have 5 tokens to use throughout the term. Tokens can be used for:

- a 48-hour extension on homework or case study due date
- revise a case study
- revise an exam

EXAM CORRECTIONS

I offer the opportunity to submit exam corrections for two reasons: (1) to correct any major misunderstandings and have a chance to revisit material that you struggled with, (2) to mitigate the impact that a bad exam day can have on your final course grade. You can submit corrections to your exam to earn back 50% of your missed points. Corrected exam grades are capped at 90%. Exam revisions are due 1 week after graded exams are returned to you, or the last day of class at 11:59pm, whichever is earlier.

CASE STUDY REVISIONS

Revised case studies are due 1 week after graded case studies are returned to you, or the last day of class at 11:59pm, whichever is earlier.

ACADEMIC INTEGRITY

You are expected to follow Carleton's policies regarding academic integrity. I encourage you to discuss the homework problems with other humans and do background reading as needed. You should code and write up your solutions on your own. Exams must be done by yourself without communicating with others; all work must be your own. You should collaborate with your teammates on projects, and should use external resources for background research and debugging, but all work should be original and all sources should be properly cited. The use of textbook solution manuals (physical or online), course materials from other students, or materials from previous versions of this course are not allowed.

Large-language models (e.g. ChatGPT, Gemini, etc.) should only be used for coding or debugging help after you've attempted to solve the problem on your own, and you should never type homework problems directly into a prompt. Copying, paraphrasing, summarizing, or submitting work generated by anyone but yourself without proper attribution is considered academic dishonesty (this includes output from LLMs).

I also have a few rules in place to protect my intellectual property. You may not record my lectures using tools such as Otter.ai or upload any video or audio recordings to generate transcripts or study notes. You may not upload my course materials (slides, assignment prompts, note sets, etc.) into AI tools or homework help sites.

"AI" tools are new for all of us and it's OK to have questions about what is and isn't appropriate. Please ask if you are unsure of whether or not your actions are complying with the assignment/quiz/project instructions. Always default to acknowledging any help received. Cases of suspected academic dishonesty are handled by the Provost's Office and I am obligated to report any suspected violations of this policy.

DIVERSITY AND INCLUSION

We all come to class with different backgrounds and experiences, and this diversity makes our class environment richer. I value diversity and inclusion, and am committed to a climate of mutual respect and full participation in and out of the classroom. This class strives to be a learning environment that is usable, equitable, inclusive and welcoming, regardless of race, ethnicity, religion, gender and gender identities, sexual orientation, ability, socioeconomic background, and nationality. If you anticipate or experience any barriers to learning, please discuss your concerns with me.

RESOURCES

PROFESSOR AVAILABILITY

Please reach out to me if you have any questions! You can reach me on Ed, email, drop-in hours, or make an appointment. Ed is typically the best venue for homework/content questions outside of drop-in hours. I'll respond to Ed messages at least 3 times each week day, and try to respond to emails within 48 hours You will receive responses faster on Ed than email. I am online irregularly over evenings and weekends to devote time to family and rest, and I hope that you also find time over the weekend to recharge and reset!

ACCOMODATIONS

Carleton College is committed to providing equitable access to learning opportunities for all students. The Office of Accessibility Resources (Henry House, 107 Union Street) is the campus office that collaborates with students who have disabilities to provide and/or arrange reasonable accommodations. If you have, or think you may have, a disability, please contact OAR@carleton.edu to arrange a confidential discussion regarding equitable access and reasonable accommodations. You are also welcome to contact me privately to discuss your academic needs. However, all disability-related accommodations must be arranged, in advance, through OAR.

STAT LAB

The Stats Lab (CMC 304) offers drop-in help R/RStudio help sessions run by friendly and knowledgeable lab assistants from 7-10pm Sunday through Thursday evenings. To make the most of your time, attempt homework problems on your own beforehand and bring your class notes and lab manual as these are helpful references for both you and the lab assistant.

TITLE IX

Please be aware that all faculty are "responsible employees", which means that if you tell me about a situation involving sexual harassment, sexual assault, dating violence, domestic violence, or stalking, I must share that information with the Title IX Coordinator. Although I have to make this notification, you will control how your case will be handled, including whether or not you wish to meet with the Title IX coordinator or pursue a formal complaint.

TAKE CARE OF YOURSELF

Do your best to maintain a healthy lifestyle this term by wearing a mask when you're sick, eating a vegetable every now and then, exercising, avoiding excessive drug and alcohol use, getting enough sleep, and taking some time to relax. Your physical and mental health is more important than your grade in this course. There are many helpful resources available on campus and an important part of the college experience is learning how to ask for help. For more information, see Student Health and Counseling (SHAC), the Office of Health Promotion, or the Office of the Chaplain. If you are experiencing physical or mental health symptoms as a result of coursework, please speak with me so we can address the problem together.